Finger MRI — 30 Sample Cases

A teaching collection of the most common finger, hand, and thumb MRI diagnoses with realistic worked dictations.

Pulleys & Flexor Tendons

Case 1 — A2 pulley tear with bowstringing (climber)

Clinical: 28-year-old male competitive rock climber, acute "pop" at the base of the ring finger during a crimp move, with pain and visible bowstringing.

> **F:** Axial and sagittal T2 fat-saturated images of the **left ring finger** show complete disruption of the **A2 pulley** at the proximal phalanx with edema and fluid tracking along the flexor sheath. On the flexed sagittal sequence, the **tendon–bone (TB) distance** at the proximal phalanx measures **5 mm** (normal <2 mm), confirming **bowstringing** of the flexor tendons. The **A3 and A4 pulleys** remain intact and the FDP/FDS tendons are continuous without retraction. No phalangeal fracture.

> **I:** Complete A2 pulley rupture with bowstringing in a climber. TB distance 5 mm supports a high-grade (Schöffl grade III–IV) injury.

Case 2 — A4 pulley tear

Clinical: 31-year-old male climber, localized middle-phalanx pain after a dynamic move; less dramatic than A2 injuries clinically.

> **F:** High-resolution axial T2 FS of the **long finger** demonstrates discontinuity of the **A4 pulley** over the middle phalanx with sheath fluid and mild adjacent soft-tissue edema. Flexed sagittal images show a **TB distance of 3 mm** at the middle phalanx indicating mild bowstringing. **A2 pulley intact.** Flexor tendons intact without tear or retraction.

> **I:** Isolated A4 pulley tear with mild bowstringing. Frequently combined with A2/A3 injury in multi-pulley patterns; isolated here.

Case 3 — Multi-pulley injury (A2 + A3 + A4)

Clinical: 35-year-old elite boulderer, loud pop while crimping, immediate swelling and inability to fully flex the finger.

> **F:** Axial and flexed sagittal T2 FS of the **ring finger** show combined rupture of the **A2, A3, and A4 pulleys** with extensive flexor sheath fluid and subcutaneous edema. The flexor tendon complex is markedly displaced volarly with a **TB distance of 7 mm**. FDP and FDS tendons are intact but bowstrung. No avulsion fracture at the proximal phalanx base.

> **I:** Multi-pulley rupture (A2/A3/A4) with marked bowstringing — Schöffl grade IV. Surgical reconstruction commonly indicated.

Case 4 — FDP avulsion (jersey finger), Leddy type II

Clinical: 22-year-old male rugby player, forced hyperextension of the flexed ring finger grabbing an opponent's jersey; cannot actively flex the DIP.

> **F:** Sagittal PD and T2 FS of the **ring finger** show avulsion of the **flexor digitorum profundus (FDP)** from its insertion on the distal phalanx base. The retracted tendon stump lies at the level of the **PIP joint (Leddy–Packer type II)**, tethered at the vinculum, with intervening hematoma along the sheath. A small **2 mm bony avulsion fragment** is noted. FDS intact.

> **I:** Jersey finger — FDP avulsion, Leddy type II with tendon retraction to the PIP joint. Retraction level guides surgical timing.

Case 5 — Flexor tendon laceration (Zone II)

Clinical: 44-year-old male chef, kitchen knife laceration to the volar long finger; unable to flex DIP and PIP.

> **F:** Axial and sagittal T2 FS through the **volar long finger** demonstrate full-thickness laceration of both **FDP and FDS tendons within Zone II** at the level of the proximal phalanx, with an **8 mm gap** between retracted tendon ends and surrounding hematoma. The overlying skin defect and sheath disruption are seen. Neurovascular bundles not clearly involved.

> **I:** Complete Zone II FDP and FDS laceration with 8 mm retraction. "No man's land" zone — primary repair with attention to both slips.

Case 6 — Stenosing flexor tenosynovitis (trigger thumb)

Clinical: 53-year-old diabetic woman, painful catching and locking of the thumb in flexion, palpable nodule at the MCP.

> **F:** Axial and sagittal PD FS of the **thumb** show thickening of the **A1 pulley** with a hypointense nodular thickening of the **flexor pollicis longus** tendon at the MCP level and a thin rim of sheath fluid. No tendon tear. Mild adjacent soft-tissue edema.

> **I:** Stenosing tenosynovitis (trigger thumb) with A1 pulley and FPL thickening. Often managed with injection or A1 release.

Case 7 — Infectious flexor tenosynovitis

Clinical: 39-year-old man, penetrating thorn injury to the volar index 4 days prior, now with fusiform swelling, fixed flexion, and pain on passive extension (Kanavel signs).

> **F:** Axial and sagittal T2 FS of the **index finger** show marked complex fluid distending the **flexor tendon sheath** with synovial thickening and avid post-contrast peripheral enhancement. Surrounding subcutaneous edema and reactive enhancement extend to the palm. The FDP/FDS tendons are intact but surrounded by debris-laden fluid. No drainable abscess or osteomyelitis.

> **I:** Pyogenic (infectious) flexor tenosynovitis. Surgical emergency — drainage and IV antibiotics indicated.

Extensor Mechanism

Case 8 — Mallet finger (terminal tendon, bony avulsion)

Clinical: 26-year-old volleyball player, ball struck the fingertip forcing flexion; cannot actively extend the DIP, fingertip droops.

> **F:** Sagittal T2 FS and PD of the **long finger** show avulsion of the **terminal extensor tendon** from the dorsal base of the distal phalanx with a **3 mm displaced bony fragment** involving roughly **30% of the articular surface**. Mild DIP joint effusion and dorsal soft-tissue edema. No volar subluxation of the distal phalanx.

> **I:** Bony mallet finger with 3 mm avulsion fragment. Fragment size/articular involvement guides splinting versus fixation.

Case 9 — Central slip rupture / boutonnière

Clinical: 34-year-old basketball player, jammed middle finger weeks ago, now with evolving fixed PIP flexion and DIP hyperextension.

> **F:** Sagittal and axial T2 FS of the **middle finger** show discontinuity of the **central slip** at its insertion on the dorsal base of the middle phalanx with surrounding edema and PIP effusion. Early **volar migration of the lateral bands** is suggested. The terminal tendon is intact.

> **I:** Central slip rupture producing a boutonnière deformity. Lateral band subluxation is the key associated finding.

Case 10 — Sagittal band rupture (boxer's knuckle)

Clinical: 29-year-old male amateur boxer, pain and "snapping" over the long-finger MCP after a punch; extensor tendon subluxes with flexion.

> **F:** Axial T2 FS at the **long finger MCP** shows disruption of the **radial sagittal band** with **ulnar subluxation of the extensor digitorum tendon** into the intermetacarpal groove on flexion. Dorsal capsular edema and a small MCP effusion. Extensor tendon itself intact.

> **I:** Radial sagittal band tear with ulnar extensor subluxation — "boxer's knuckle." Radial band tears typically cause ulnar tendon dislocation.

Case 11 — Extensor tendon laceration (dorsal, Zone VI)

Clinical: 41-year-old man, dorsal hand laceration from broken glass; loss of MCP extension of the index.

> **F:** Axial and sagittal T2 FS over the **dorsal index** at the metacarpal level (**Zone VI**) show full-thickness laceration of the **extensor digitorum tendon** with a **6 mm gap** between tendon ends and adjacent hematoma. The juncturae tendinum partially maintains alignment. No fracture.

> **I:** Zone VI extensor tendon laceration with 6 mm gap. Junctura may mask the deficit clinically.

Ligaments & Volar Plate

Case 12 — Thumb UCL tear with Stener lesion (skier's thumb)

Clinical: 38-year-old skier, fell on an abducted thumb with a planted pole; pain and instability at the thumb MCP with valgus stress.

> **F:** Coronal T2 FS of the **thumb MCP** shows a full-thickness tear of the **ulnar collateral ligament** at its distal (phalangeal) attachment, with the retracted, balled-up ligament displaced **proximal and superficial to the adductor aponeurosis** — a **Stener lesion** ("yo-yo on a string"). Adjacent edema and a small MCP effusion. No fracture.

> **I:** Complete thumb UCL tear with Stener lesion. Surgical repair required — the interposed aponeurosis prevents healing.

Case 13 — Thumb UCL tear without Stener lesion

Clinical: 45-year-old woman, fall onto outstretched abducted thumb; MCP pain, mild laxity.

> **F:** Coronal and axial T2 FS of the **thumb MCP** show a full-thickness tear of the **UCL** at the distal attachment, but the torn ligament remains **deep to and beneath the adductor aponeurosis** in near-anatomic position (no Stener). Surrounding edema and small effusion. Volar plate and RCL intact.

> **I:** Complete thumb UCL tear without Stener lesion. Non-displaced fibers favor conservative or surgical management depending on stability.

Case 14 — Thumb radial collateral ligament (RCL) injury

Clinical: 33-year-old man, forced adduction injury of the thumb; tenderness over the radial MCP.

> **F:** Coronal T2 FS of the **thumb MCP** shows a partial-thickness tear of the **radial collateral ligament** with thickening, intrasubstance high signal, and surrounding edema; fibers remain in continuity. Small MCP effusion. UCL and volar plate intact.

> **I:** Partial RCL tear of the thumb MCP. Less common than UCL injury; usually managed conservatively.

Case 15 — PIP volar plate injury (dorsal dislocation)

Clinical: 24-year-old man, hyperextension "jam" of the index PIP during a fall; swelling and dorsal tenderness.

> **F:** Sagittal T2 FS of the **index PIP** shows disruption of the **volar plate** at its distal attachment on the middle phalanx base with a small **2 mm avulsion fragment** and surrounding edema. PIP effusion present. Collateral ligaments intact; no persistent dislocation.

> **I:** Volar plate injury with small avulsion at the middle phalanx base, consistent with a reduced dorsal PIP dislocation.

Case 16 — Collateral ligament tear of a finger PIP

Clinical: 30-year-old woman, lateral "jamming" force to the small finger PIP; localized radial-sided pain and instability.

> **F:** Coronal T2 FS of the **small finger PIP** shows a full-thickness tear of the **radial collateral ligament** at its proximal-phalanx origin with retraction and fluid in the gap. Adjacent edema and effusion. Volar plate intact.

I: Complete radial collateral ligament tear of the small finger PIP. Stress instability correlates clinically.

Case 17 — Chronic gamekeeper's thumb

Clinical: 58-year-old farmer, chronic ulnar-sided thumb pain and weak pinch over many months.

> **F:** Coronal T2 FS and PD of the **thumb MCP** show a thickened, irregular, attenuated **UCL** with chronic intrasubstance signal, no acute edema, and a small **ossific fragment** at the phalangeal attachment. Mild MCP joint-space narrowing and early osteophytes from chronic instability.

> **I:** Chronic gamekeeper's thumb — attenuated UCL with secondary degenerative change. Distinct from acute skier's thumb.

Bone, Joint & Other

Case 18 — Condylar fracture of the proximal phalanx

Clinical: 19-year-old man, axial/lateral load to the long finger in a fall; localized PIP swelling and tenderness.

> **F:** Coronal and sagittal PD/T2 FS of the **long finger** show a **unicondylar fracture** of the proximal phalangeal head with a **2 mm articular step-off** at the PIP and surrounding marrow edema. Small PIP effusion. Collateral ligaments and volar plate intact.

> **I:** Unicondylar proximal phalanx fracture with 2 mm articular step-off. Intra-articular involvement warrants orthopedic evaluation for fixation.

Case 19 — PIP/DIP osteoarthritis

Clinical: 64-year-old woman, chronic finger stiffness and bony enlargement (Heberden and Bouchard nodes).

> **F:** PD and T2 FS of the **index and long fingers** show **DIP and PIP joint-space narrowing**, marginal **osteophytes**, subchondral cysts, and subchondral marrow edema-like signal. Small effusions without erosive synovitis or pannus. No marginal erosions.

> **I:** Primary osteoarthritis of the DIP (Heberden) and PIP (Bouchard) joints. Absence of erosive pannus distinguishes from inflammatory arthritis.

Case 20 — Rheumatoid arthritis (erosions/pannus)

Clinical: 47-year-old woman, symmetric MCP/PIP pain, morning stiffness, positive RF/anti-CCP.

> **F:** Coronal post-contrast T1 FS of the hand show **enhancing synovial pannus** distending the **2nd and 3rd MCP joints** with marginal **erosions** (largest ~**4 mm** at the 2nd metacarpal head), subchondral marrow edema (osteitis), and tenosynovial enhancement of the extensor compartments. No DIP involvement.

> **I:** Inflammatory arthritis with enhancing pannus, marginal erosions, and osteitis — consistent with rheumatoid arthritis. MCP/PIP predominance with DIP sparing is typical.

Case 21 — Gout with tophi

Clinical: 56-year-old man, recurrent painful swelling of the fingers, hyperuricemia.

> **F:** PD and T2 FS of the **index DIP** show a juxta-articular soft-tissue mass of **intermediate-to-low T2 signal** (**tophus**) with adjacent **"punched-out" erosions** with sclerotic, overhanging margins and preserved joint space until late. Mild surrounding edema. No diffuse synovial pannus.

> **I:** Tophaceous gout with characteristic punched-out periarticular erosions and intermediate-signal tophi. Relative joint-space preservation distinguishes from RA.

Case 22 — Glomus tumor (subungual)

Clinical: 42-year-old woman, exquisite point tenderness and cold sensitivity under the fingernail for over a year.

> **F:** Axial and sagittal T2 FS and post-contrast T1 FS of the **index fingertip** show a small **6 mm** well-circumscribed **subungual** nodule that is **markedly T2 hyperintense** with **intense homogeneous enhancement**, with subtle scalloping of the adjacent dorsal distal phalangeal cortex. No aggressive features.

> **I:** Subungual glomus tumor. Classic triad of pain, cold sensitivity, and point tenderness with an avidly enhancing T2-bright nodule.

Case 23 — Digital mucoid (ganglion) cyst at DIP

Clinical: 60-year-old woman, dorsal DIP swelling with longitudinal nail grooving and underlying osteoarthritis.

> **F:** Sagittal and axial T2 FS of the **long finger** show a well-defined **lobulated T2-hyperintense cyst** (about **8 mm**) arising from the **dorsal DIP joint** with a stalk communicating with the joint, abutting the germinal nail matrix. Adjacent DIP osteophytes and joint-space narrowing. Thin peripheral enhancement only.

> **I:** Digital mucoid (ganglion) cyst arising from an osteoarthritic DIP, accounting for the nail deformity.

Case 24 — Giant cell tumor of the tendon sheath

Clinical: 37-year-old woman, slowly growing painless firm nodule on the volar finger.

> **F:** Axial and sagittal T1, T2 FS, and gradient-echo of the **volar long finger** show a well-defined lobulated soft-tissue mass (**14 mm**) adjacent to the **flexor tendon sheath**, **low-to-intermediate on T1 and T2** with characteristic **blooming on gradient-echo** from hemosiderin, and avid enhancement. Mild scalloping of the adjacent phalanx without marrow invasion.

> **I:** Giant cell tumor of the tendon sheath (localized PVNS). Hemosiderin blooming and tendon-sheath origin are characteristic.

Case 25 — Osteomyelitis of the distal phalanx

Clinical: 61-year-old diabetic man, chronic fingertip ulcer with discharge and persistent swelling.

> **F:** T1, T2 FS, and post-contrast T1 FS of the **distal phalanx of the long finger** show **confluent low T1 / high T2 marrow signal** with avid enhancement, cortical destruction, and an adjacent enhancing soft-tissue collection with overlying skin ulceration consistent with a **rim-enhancing abscess**. Findings indicate **osteomyelitis** with septic involvement of the adjacent DIP.

> **I:** Osteomyelitis of the distal phalanx with adjacent soft-tissue abscess and septic DIP arthritis. Marrow T1 replacement is the most reliable sign.

Case 26 — Dupuytren contracture

Clinical: 66-year-old man of Northern European descent, painless palmar nodule with progressive flexion contracture of the ring finger.

> **F:** Axial and sagittal T1 and T2 FS of the **palm** show a **hypointense (T1/T2) nodular and cord-like thickening** of the **palmar aponeurosis** extending from the distal palm toward the ring-finger MCP, superficial to the flexor tendons which are normal. Low signal reflects fibrous/collagenous tissue; minimal enhancement suggests a quiescent stage.

> **I:** Dupuytren contracture — palmar fascial cord and nodule. Low T2 signal indicates mature fibrosis with lower recurrence/progression risk.

Case 27 — FDP avulsion, Leddy type I (retraction to palm)

Clinical: 25-year-old male football player, jersey-finger mechanism with severe pain and palmar swelling.

> **F:** Sagittal and axial T2 FS of the **ring finger** show **FDP avulsion** from the distal phalanx with the tendon stump **retracted into the palm (Leddy–Packer type I)**, indicating disruption of both vincula and compromised vascular supply, with a fluid-filled empty distal sheath. No bony fragment.

> **I:** Jersey finger, Leddy type I with retraction into the palm. Type I requires urgent repair due to devascularization.

Case 28 — Pulp-space felon with early osseous involvement

Clinical: 48-year-old woman, throbbing pulp-space infection of the thumb tip after a splinter, with tense swelling.

> **F:** Axial and sagittal post-contrast T1 FS of the **thumb tip** show a **rim-enhancing collection** within the septated pulp space (**felon**) with surrounding cellulitic enhancement and early reactive marrow edema in the distal phalangeal tuft without frank cortical destruction.

> **I:** Pulp-space felon with early reactive osteitis. Drainage indicated to prevent progression to distal phalanx osteomyelitis.

Case 29 — Chronic boutonnière with lateral band subluxation

Clinical: 52-year-old woman with longstanding rheumatoid arthritis, fixed boutonnière deformity of the long finger.

> **F:** Sagittal and axial T2 FS of the **long finger** show chronic attenuation/disruption of the **central slip** with **volar subluxation of the lateral bands** below the PIP axis of rotation, fixed PIP flexion, and compensatory DIP hyperextension. Associated synovitis and small marginal erosions reflect the underlying RA.

> **I:** Chronic boutonnière deformity with volar lateral band subluxation, secondary to rheumatoid central slip attrition.

Case 30 — Chronic volar plate injury with PIP osteoarthritis

Clinical: 55-year-old man with prior PIP hyperextension injury, now chronic stiffness and a small swan-neck tendency.

> **F:** Sagittal PD and T2 FS of the **index PIP** show chronic thickening and scarring of the **volar plate** with a small ossific focus at the middle phalanx base, mild PIP joint-space narrowing, marginal osteophytes, and subchondral cysts. No acute edema or effusion.

> **I:** Chronic volar plate injury with secondary PIP osteoarthritis. Post-traumatic degenerative change from prior hyperextension injury.